



POLITECNICO
MILANO 1863

ASSIGNMENT 1

STRUCTURAL DYNAMICS AND AEROELASTICITY

Images and Comments

POLITECNICO DI MILANO

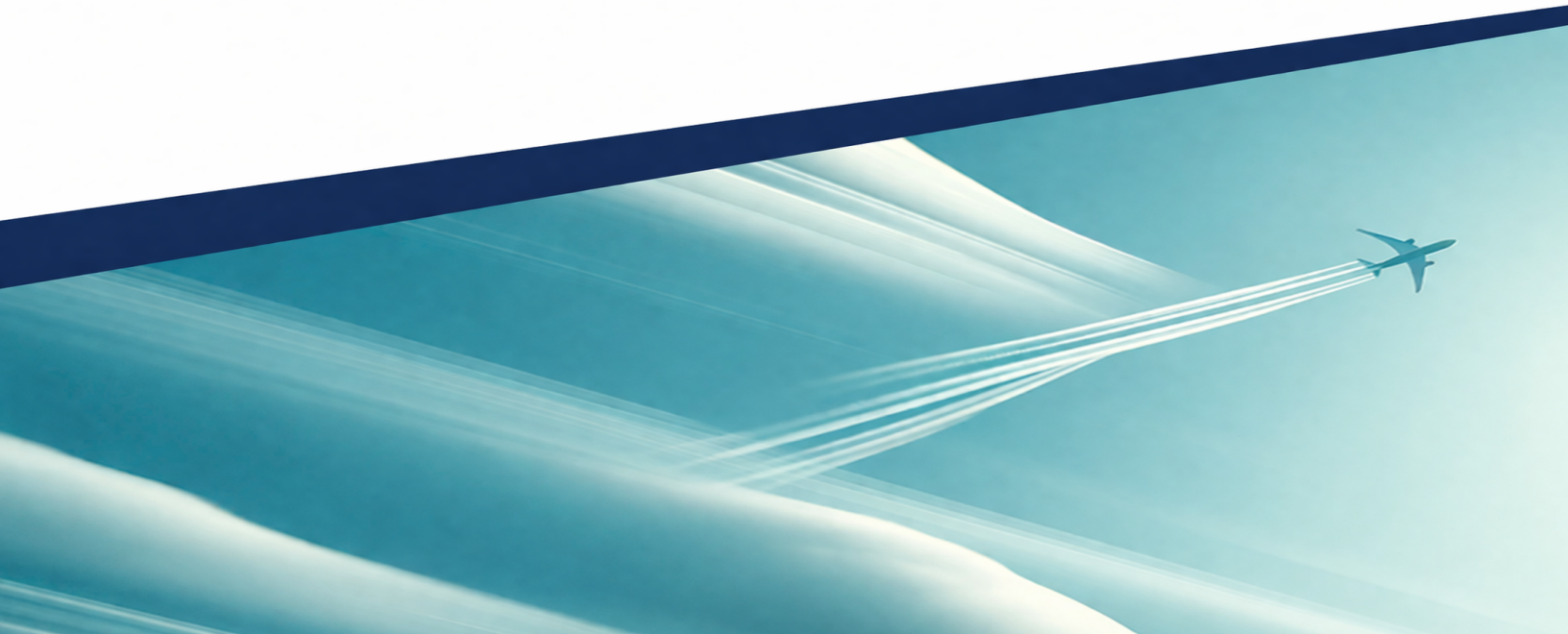
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Influence of strut position on divergence

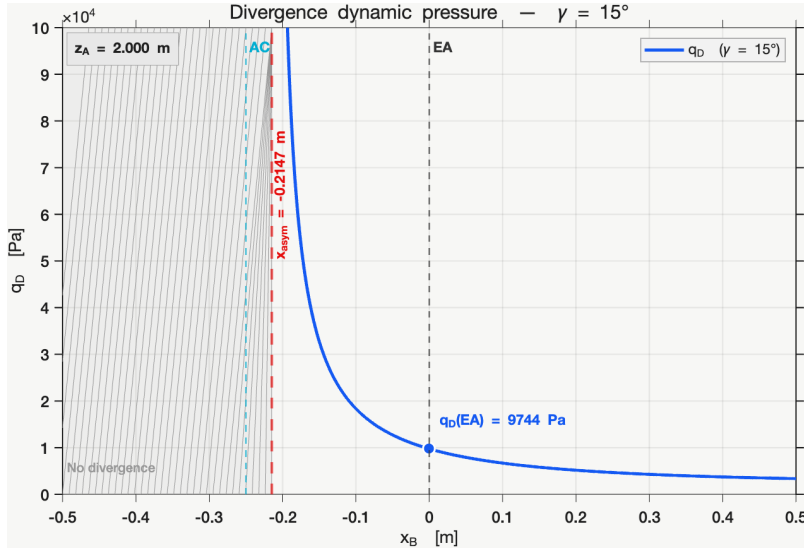


FIGURE 1 – q_D vs. x_B for $\gamma = 15^\circ$.

q_D is highly sensitive to x_B . A vertical asymptote at $x_B \simeq -0.2147$ m separates two regimes :

- **Right** : divergence at positive q_D , dropping rapidly toward the elastic axis ($q_D \approx 9.7 \times 10^3$ Pa).
- **Left** : flexural-torsional stabilization suppresses static divergence entirely.

Limitations. No-divergence region holds within the linear model only – flutter and control reversal remain unconstrained. Forward strut placement may incur buckling/mass penalties and is sensitive to geometric tolerances near the asymptotic boundary.

Influence of γ on the control effectiveness

x_B fixed; $E_C = L/L_R$ with $E_C = 1$ rigid limit, $E_C > 1$ amplification, $E_C < 0$ reversal. All curves start at $E_C = 1$. Three regimes :

- **Low** γ ($10^\circ, 12.5^\circ$) : $E_C > 1$ – elastic twist amplifies aileron effect – then blocked by divergence; displayed value is actually q_D , not q_C .
- **Intermediate** (15°) : monotonic decrease, genuine reversal at $q_C = 29702$ Pa.
- **High** γ ($17.5^\circ, 20^\circ$) : detrimental coupling from low q ; $q_C = 24685$ and 21427 Pa – reversal pressure decreases with γ .

$E_C \sim [\det \mathbf{K}_{TOT}]^{-1}$; negative branch past divergence is a mathematical artefact. γ controls sign and magnitude of aeroelastic feedback : low $\gamma \rightarrow$ favorable amplification (divergence governs); high $\gamma \rightarrow$ early control reversal.

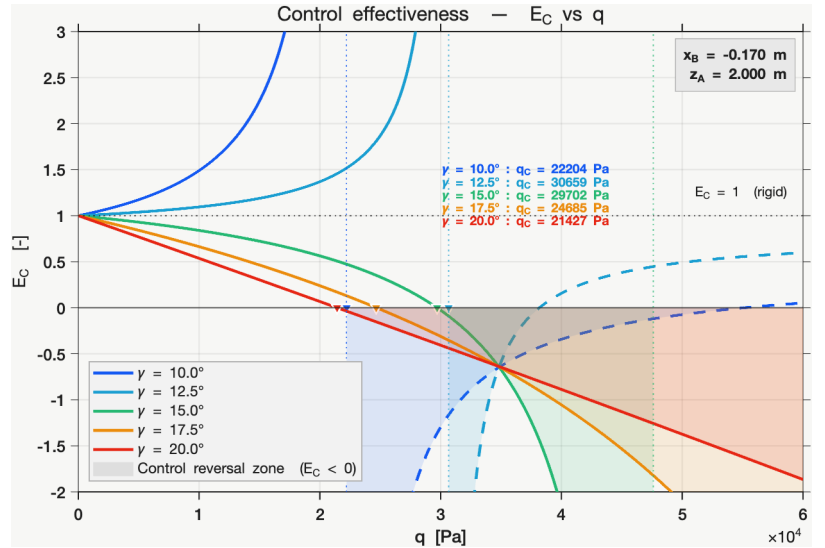
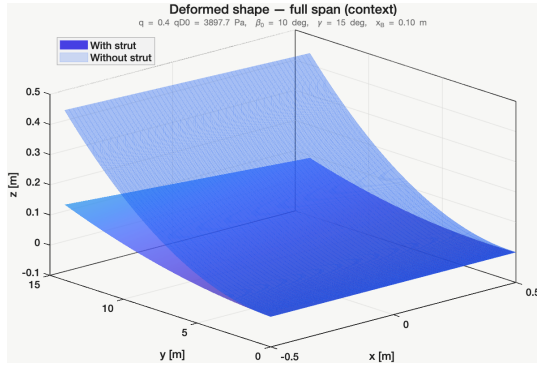


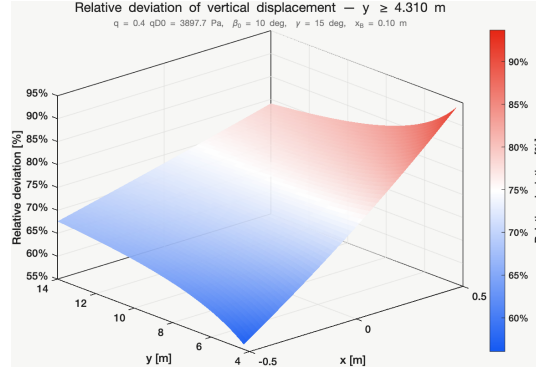
FIGURE 2 – E_C vs. q for several γ , x_B fixed.

Comparison of the deformed shape with and without the strut

Config : $x_B = 0.10$ m, $\beta_0 = 10^\circ$, $q = 0.4 q_{D,0}$ (subcritical – differences attributable to strut stiffening, not instability proximity). δ_z interpreted only for $y \geq y_{\min} = 4.31$ m ($\approx 30.8\% b$), defined as the first station where the absolute gap at $x = c/4$ reaches 10% of its tip value, to avoid the root singularity ($z_{ns} \rightarrow 0$).



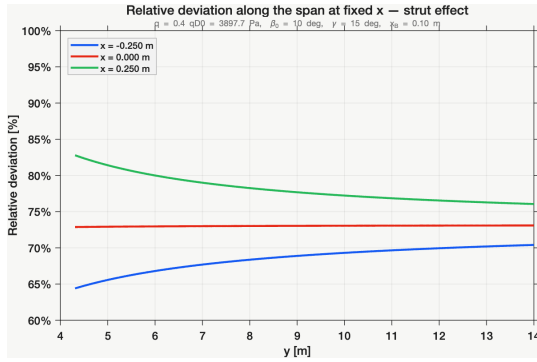
3D shape, with vs. without strut.



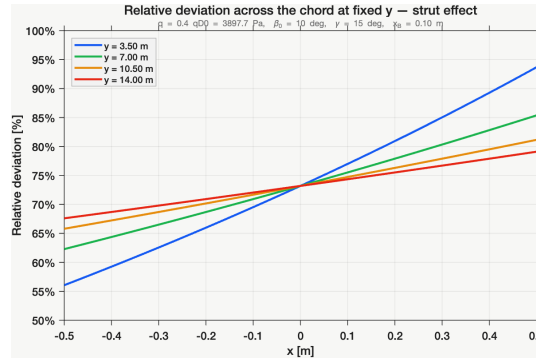
δ_z map, $y \geq y_{\min}$.

Both configurations share the same bending–torsion pattern (zero at root, increasing toward tip); the strut reduces amplitude without altering shape :

1) $\delta_z \in \sim [50\%, 90\%]$ over valid span – strut wing is $\frac{1}{4}-\frac{1}{3}$ as deformed as no-strut reference.



Spanwise cuts of δ_z .



Chordwise cuts of δ_z .

2) Along elastic axis ($x = 0$) : $z = (y/b)^2 \phi_w$, $\delta_z \approx 70\%$ – dominant bending reduction.

3) Chordwise variation of δ_z consistent with torsional term $-x(y/b)\phi_\theta$ – strut simultaneously stiffens bending and torsion.